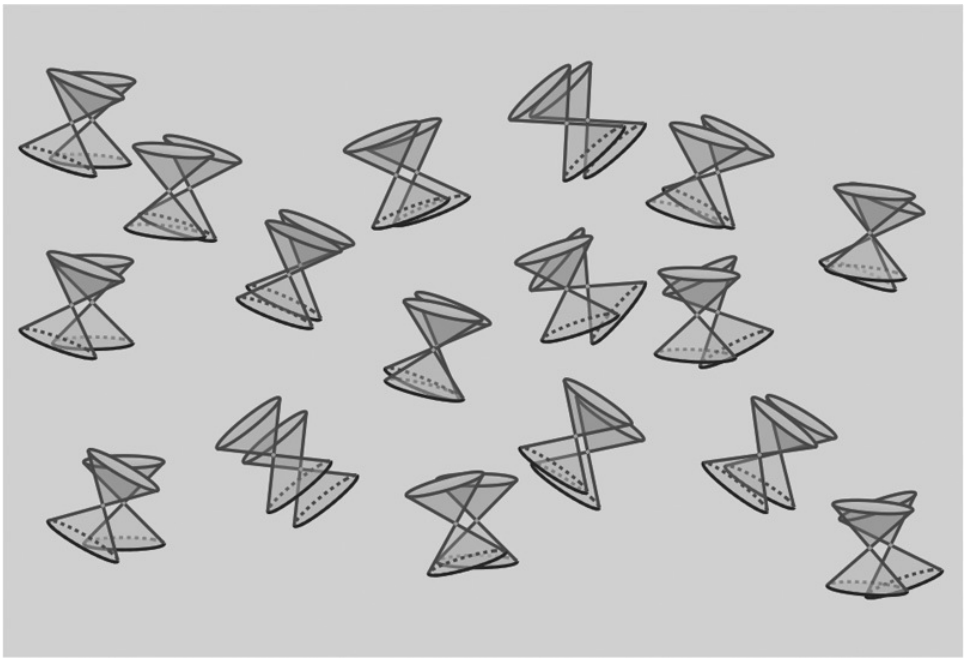


Even the distinction between present, past, and future thus becomes fluctuating, indeterminate. Just as a particle may be diffused in space, so, too, the differences between past and future may fluctuate: an event may be both before and after another one.

RELATIONS

"Fluctuation" does not mean that what happens is *never* determined. It means that it is determined only at certain moments, and in an unpredictable way. Indeterminacy is resolved when a quantity interacts with something else.*

In the interaction, an electron materializes at a certain point. For example, it collides with a screen, is captured by a particle detector, or collides with a photon—thus acquiring a concrete position.

But there is a strange aspect to this materialization of the electron: the electron is concrete only *in relation to* the other physical objects it is interacting with.[56] With regard to all the others, the effect of the interaction is only to spread the contagion of indeterminacy. Concreteness occurs only in relation to a physical system: this, I believe, is the most radical discovery made by quantum mechanics.*

When an electron collides with an object—the screen of an old television set with a cathode ray tube, for example—the cloud of probability with which we conceived of it "collapses" and the electron materializes at a point on the screen, producing one of the luminous dots that goes into the making of a TV image. But it is only in relation to the screen that this happens. In relation to